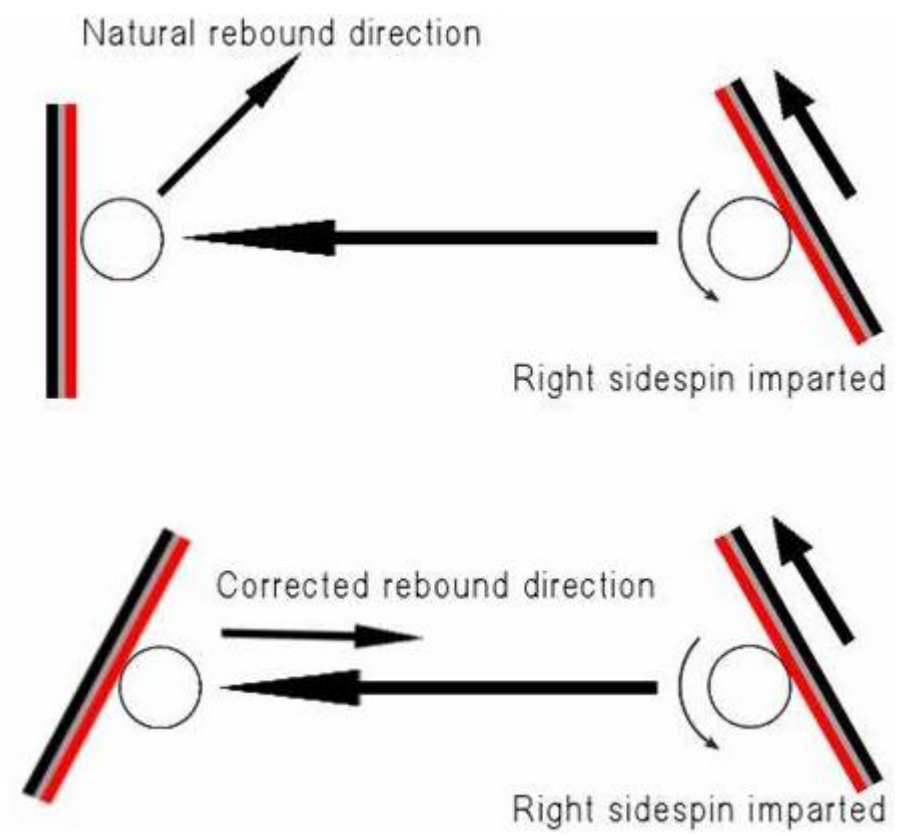
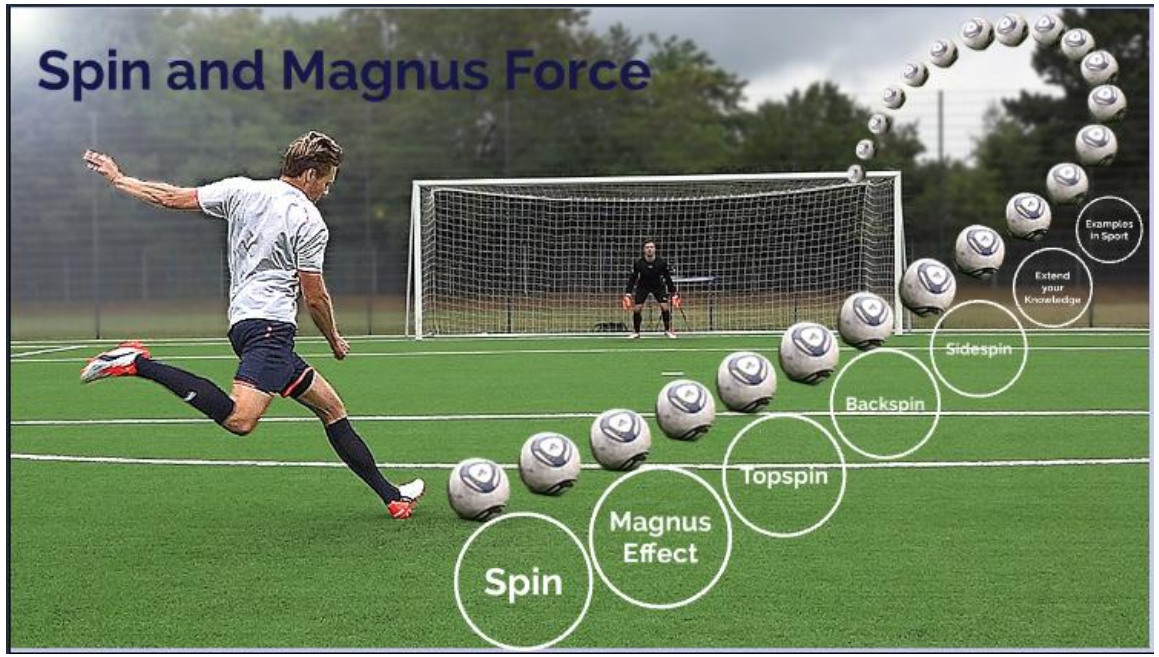
北京大学
PEKING UNIVERSITY

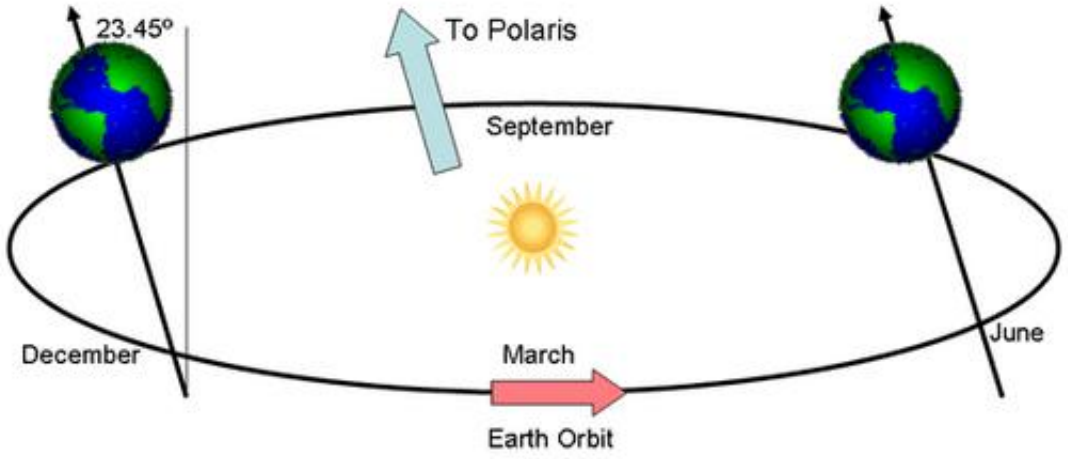
第二章 原子结构与光谱

2. 多电子原子

一　自旋



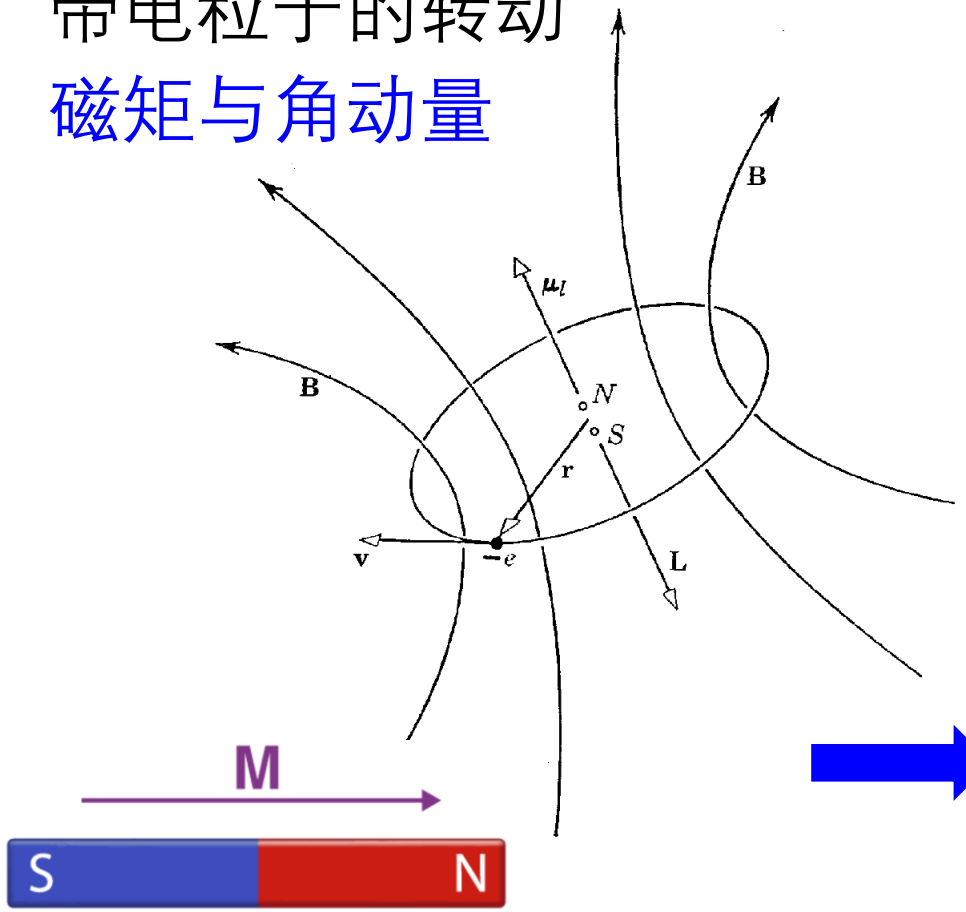




spin vs orbit

带电粒子的转动

磁矩与角动量



$$i = \frac{e}{T} = \frac{ev}{2\pi r}$$ 环形电流

轨道磁矩（orbital magnetic dipole moment）

$$\mu_l = iA$$

$$= \frac{ev}{2\pi r}\pi r^2 = \frac{evr}{2}$$

$$L = mvr$$ 轨道角动量

$$\frac{\mu_l}{L} = \frac{evr}{2mvr} = \frac{e}{2m}$$

量子问题改写为

$$\frac{\mu_l}{L} = \frac{g_l \mu_b}{\hbar}$$

引入玻尔磁子 (Bohr magneton) $\mu_b = \frac{e\hbar}{2m}$

考虑到方向

$$\boldsymbol{\mu}_l = -\frac{g_l \mu_b}{\hbar}\mathbf{L}$$

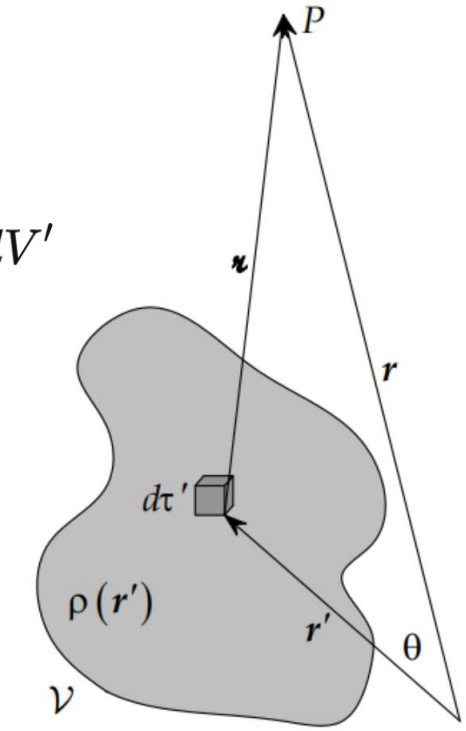
$$g_l = 1$$

静电势多级展开（multipole expansion）

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0 r}\int_{V'} \rho(\mathbf{r}')\left(1 + \frac{\hat{\mathbf{r}}\cdot\mathbf{r}'}{r} + \frac{1}{2r^2}\left(3(\hat{\mathbf{r}}\cdot\mathbf{r}')^2 - r'^2\right) + O\left(\frac{r'}{r}\right)^3\right) dV'$$

$$V(\mathbf{r}) = V_{\text{mon}}(\mathbf{r}) + V_{\text{dip}}(\mathbf{r}) + V_{\text{quad}}(\mathbf{r}) + \dots$$

$$V(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\left[\frac{q}{r} + \frac{\boldsymbol{p}\cdot\boldsymbol{r}}{r^3} + \frac{1}{2}\sum_{ij} Q_{ij}\frac{x_i x_j}{r^5}\right]$$



$$\boldsymbol{p} = \int \boldsymbol{r}'\rho(\boldsymbol{r}')d\boldsymbol{r}'$$

磁矢势多级展开

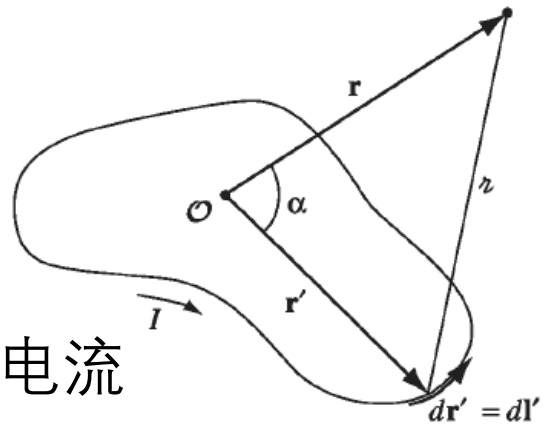
$$\boldsymbol{A}(\boldsymbol{r}) = \frac{\mu_0}{4\pi}\left[\frac{\boldsymbol{\mu}\times\boldsymbol{r}}{r^3} + \mathcal{O}\left(\frac{1}{r^3}\right)\right]$$

$$\boldsymbol{\mu} = \frac{1}{2}\int \boldsymbol{r}'\times\boldsymbol{j}(\boldsymbol{r}')d\boldsymbol{r}'$$

没有 $\frac{1}{r}$ 项（磁单极不存在）

$$\boldsymbol{A} = \frac{1}{2}\oint \boldsymbol{r}'\times d\boldsymbol{l}'$$

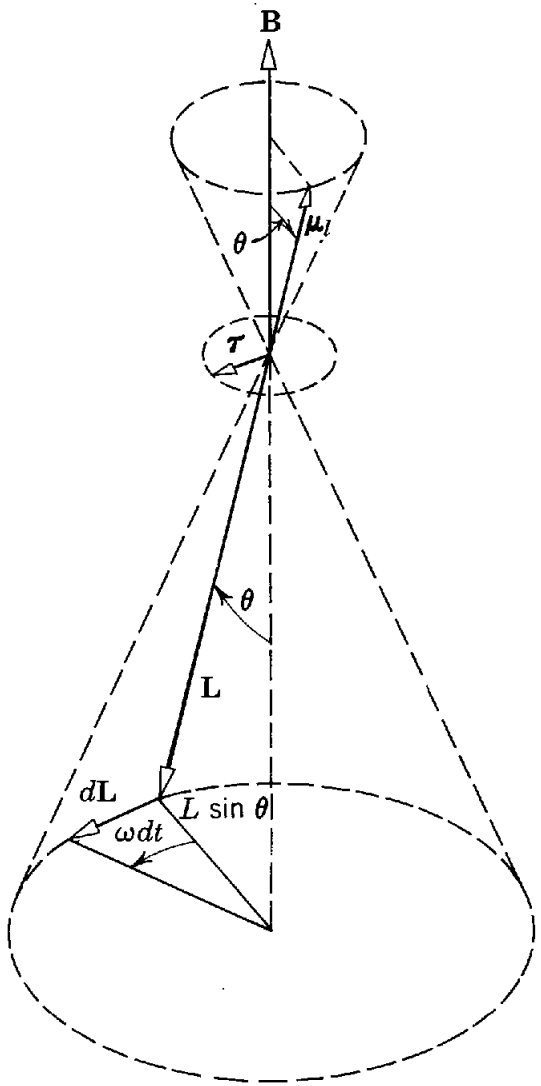
vector area of the loop



环形电流

$$\boldsymbol{\mu} = I\boldsymbol{A}$$

小磁体（磁矩）在**匀强磁场**里受力



$$\boldsymbol{\tau} = \boldsymbol{\mu}_l \times \mathbf{B}$$

$$\Delta E = -\boldsymbol{\mu}_l \cdot \mathbf{B}$$

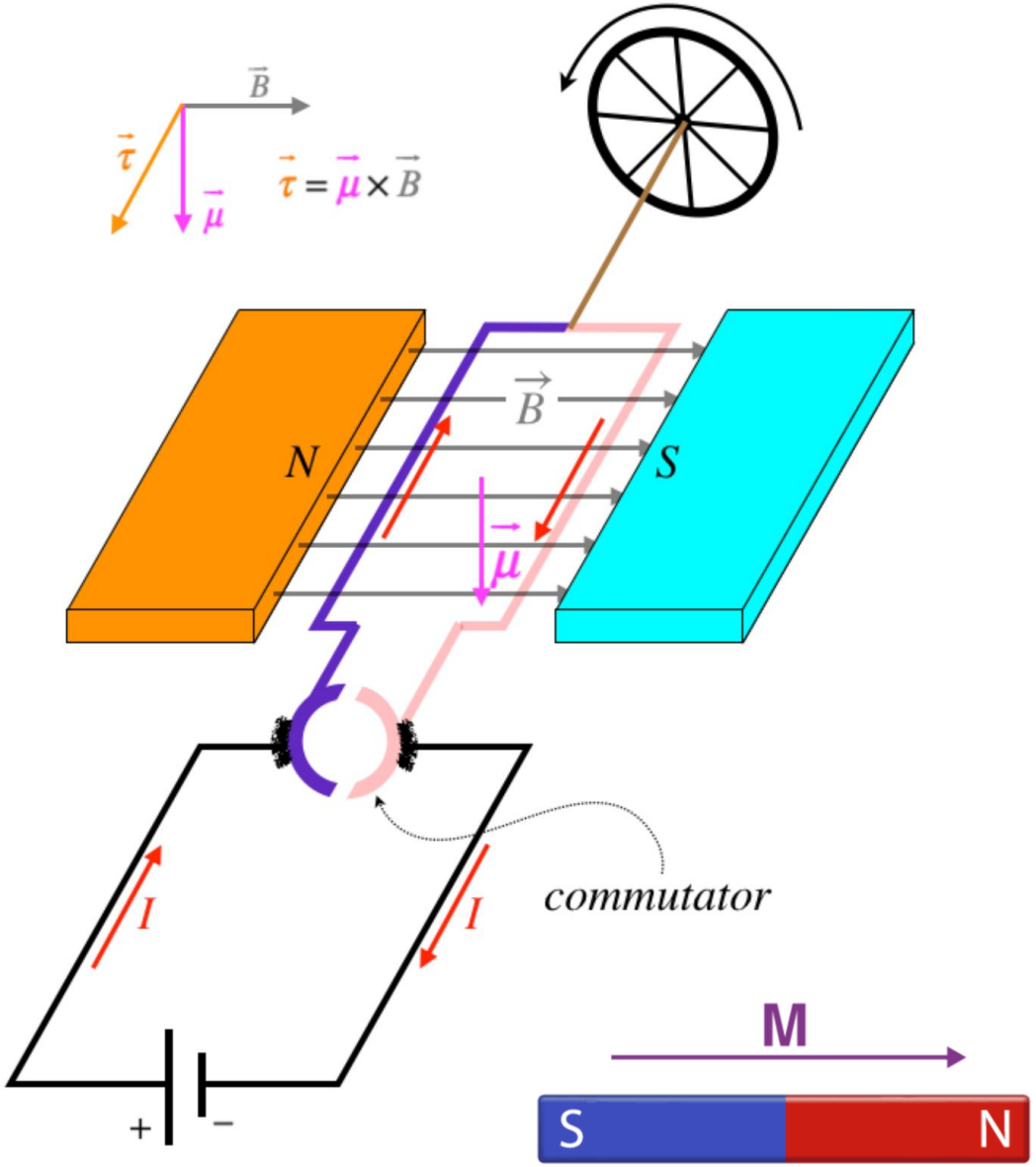
磁矩与磁场平行时，能量最低

不平行，则受力进动

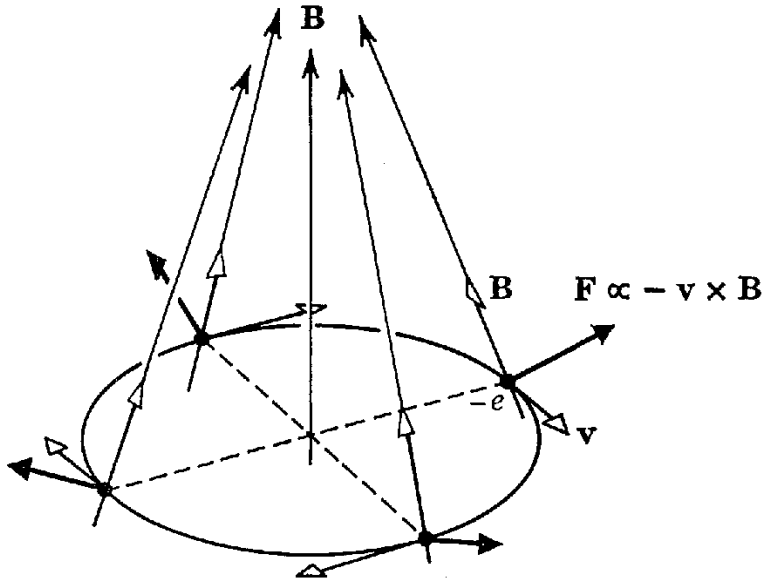
Larmor precession

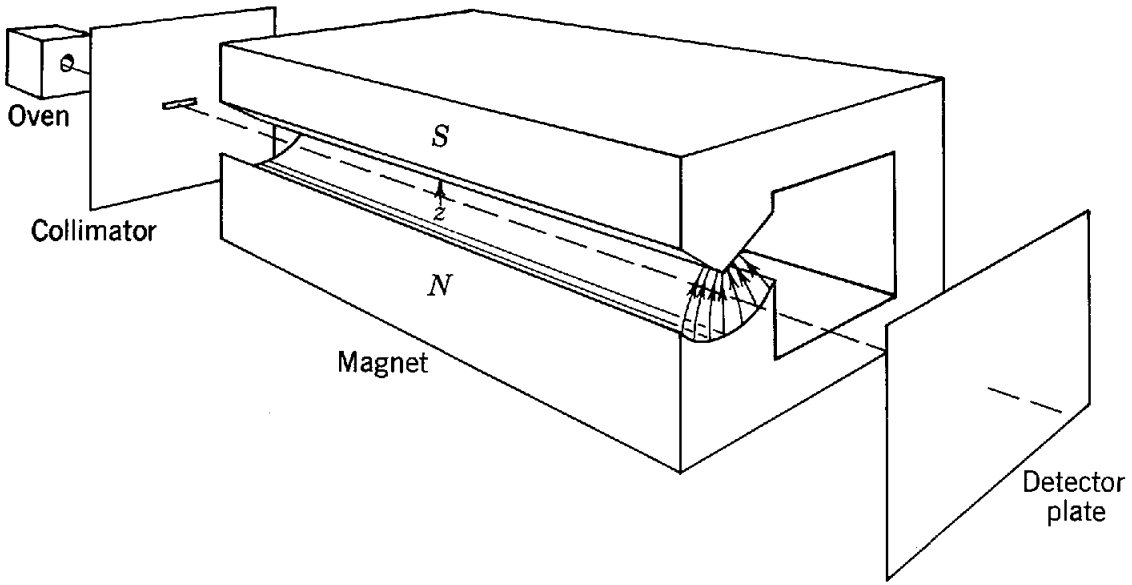
$$\boldsymbol{\omega} = \frac{g_l \mu_b}{\hbar} \mathbf{B}$$

$$L\omega \sin\theta = dL/dt = \tau = (g_l \mu_b/\hbar) LB \sin\theta$$



电子在**非匀强磁场**里的受力



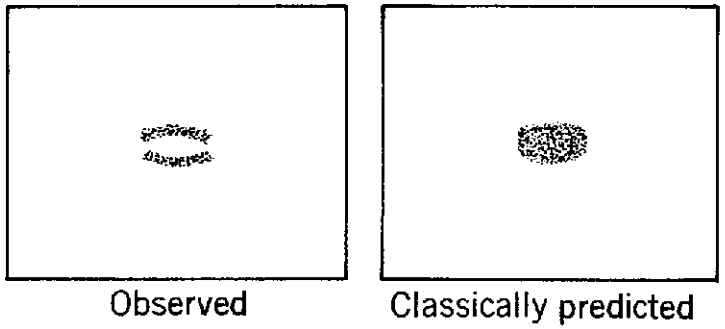


Stern-Gerlach (Ag atom, 1921)

& Phipps-Taylor (H atom, 1927)

向上会聚的磁场，施加向上的力

平均受力大小 $\overline{F_z} = \frac{\partial B_z}{\partial z} \mu_{l_z}$



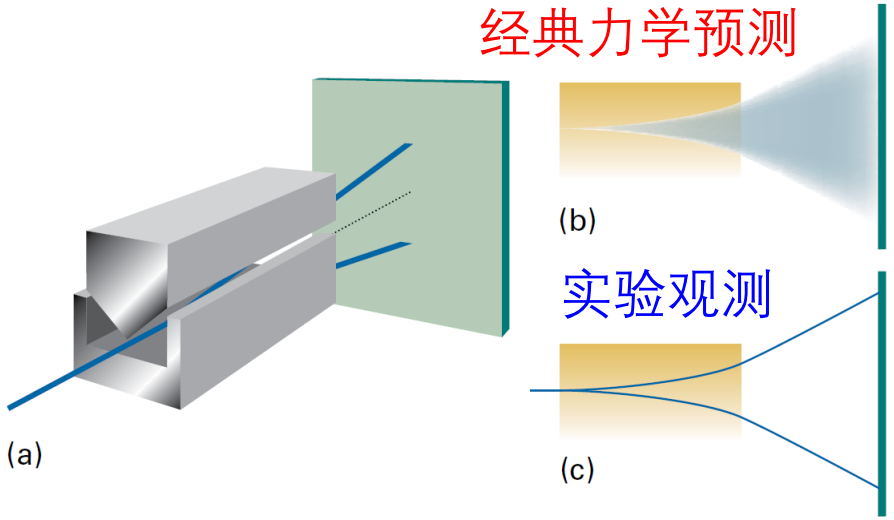
自旋角动量

- 银原子束通过非匀强磁场散射

- 电子绕核运动具有角动量
  相当于小磁体，在磁场中会发生偏转

- 实验结果，只有两种偏转模式
  - 角动量量子数为半整数

- 定义为电子自旋，$s = 1/2$

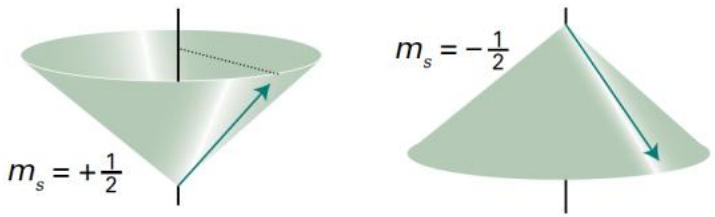


斯特恩-盖拉赫实验 (1921)

$$S = \hbar\sqrt{s(s+1)}$$

自旋是一种特殊的旋转，
没有经典对应

S: 自旋角动量的模
s: 自旋量子数
m_s: 自旋磁量子数



电子自旋的故事

1921 Stern-Gerlach银原子束实验

$$\boldsymbol{\mu}_s = -\frac{g_s \mu_b}{\hbar}\mathbf{S}$$

$$\mu_{s_z} = -g_s \mu_b m_s$$

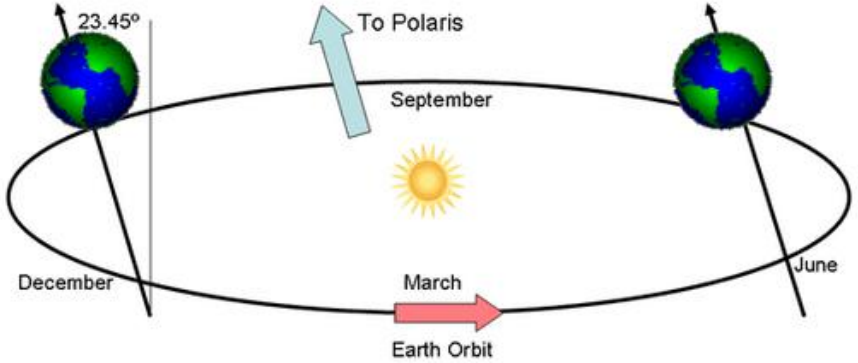
$$g_s = 2$$

1921 Compton猜测电子像陀螺一样自旋

1925 Uhlenbeck-Goudsmit正式提出电子自旋的概念

I have already sent your letter in long ago; you are both young enough to allow yourselves some foolishness!

-- Ehrenfest



1927 Phipps-Taylor氢原子束实验

1929 Dirac提出相对论量子力学，完美解释自旋

电子是质点，没有半径!

经典图像不正确

电子的自旋状态表示

$$m_s = \frac{1}{2} \Longleftrightarrow \alpha \Longleftrightarrow \uparrow$$

$$m_s = -\frac{1}{2} \Longleftrightarrow \beta \Longleftrightarrow \downarrow$$

自旋半整数的叫做费米子(Fermions)

自旋整数的叫做玻色子(Bosons)

多电子波函数满足**交换对称性**

$$\Psi_F(1,2) = -\Psi_F(2,1)$$

$$\Psi_B(1,2) = \Psi_B(2,1)$$

每个粒子的维度 = 空间 + 自旋

$$\boldsymbol{x} = (\boldsymbol{r}, m_s)$$

其他粒子(包括原子核)的自旋

$s=0$	^{12}C ^{16}O
$s=\frac{1}{2}$	电子、质子、中子、^{13}C ^{19}F
$s=1$	光子、氘、^{14}N
$s=\frac{3}{2}$	^{11}B
$s=\frac{5}{2}$	^{17}O

质子、中子都是自旋1/2粒子，但质量比电子大很多，显得转得慢

$$S = \hbar\sqrt{s(s+1)} \quad s = 0, \frac{1}{2}, 1, \frac{3}{2}, \cdots$$

$$m_s = -s, -s+1, \cdots, s$$

二　交换对称性

全同粒子假设：量子力学中，全同粒子不可分辨

意味着对任何可观测量Ω，

$$\langle\Psi|\hat{\Omega}|\Psi\rangle = \langle\hat{P}\Psi|\hat{\Omega}|\hat{P}\Psi\rangle$$

$$\forall\Psi = \Psi(\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_N) \in \mathcal{H}$$

置换算符 $\hat{P} : \mathcal{H} \longrightarrow \mathcal{H}$

$$\hat{P}\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_N) = \Psi(\boldsymbol{x}_{P(1)}, \boldsymbol{x}_{P(2)}, \cdots, \boldsymbol{x}_{P(N)})$$

可以证明：

$$[\hat{P}, \hat{\Omega}] = 0$$

进而可证：算符满足置换对称性

$$\hat{\Omega}(\boldsymbol{x}_{P(1)}, \boldsymbol{x}_{P(2)}, \cdots, \boldsymbol{x}_{P(N)}) = \Omega(\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_N)$$

经典粒子可以分辨出**运动轨迹**，量子不行！

思考：多电子波函数所在的希尔伯特空间与单电子波函数所在的空间是什么关系？

先证明置换算符的酉正性(unitary)

计算**任意**波函数Φ和Ψ的内积

$$\underline{\underline{\boldsymbol{x}}} = (\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_N)$$

$$\int d\underline{\underline{\boldsymbol{x}}} = \int d\boldsymbol{x}_1 \int d\boldsymbol{x}_2 \cdots \int d\boldsymbol{x}_N$$

$$\int d\boldsymbol{x} \equiv \sum_{m_s} \int d\boldsymbol{r}$$

$$\begin{aligned}
\langle\Phi|\Psi\rangle &= \int \langle\Phi|\underline{\underline{\boldsymbol{x}}}\rangle d\underline{\underline{\boldsymbol{x}}} \langle\underline{\underline{\boldsymbol{x}}}|\Psi\rangle \\
&= \int d\underline{\underline{\boldsymbol{x}}} \Phi^*(\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_N)\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_N) \\
&= \int d\underline{\underline{\boldsymbol{x}}} \Phi^*(\boldsymbol{x}_{P(1)}, \boldsymbol{x}_{P(2)}, \cdots, \boldsymbol{x}_{P(N)})\Psi(\boldsymbol{x}_{P(1)}, \boldsymbol{x}_{P(2)}, \cdots, \boldsymbol{x}_{P(N)}) \\
&= \int d\underline{\underline{\boldsymbol{x}}} \left[\hat{P}\Phi^*(\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_N)\right] \hat{P}\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_N) \\
&= \int \langle\hat{P}\Phi|\underline{\underline{\boldsymbol{x}}}\rangle d\underline{\underline{\boldsymbol{x}}} \langle\underline{\underline{\boldsymbol{x}}}|\hat{P}\Psi\rangle = \langle\hat{P}\Phi|\hat{P}\Psi\rangle = \langle\Phi|\hat{P}^\dagger\hat{P}\Psi\rangle
\end{aligned}$$

从而

$$\hat{I} = \hat{P}^\dagger\hat{P} \qquad \hat{P}^{-1} = \hat{P}^\dagger$$

置换算符是酉正算符

再证明 $[\hat{P},\hat{\Omega}]=0$

$$\forall\Psi\in\mathcal{H}\qquad \langle\Psi|\hat{\Omega}|\Psi\rangle=\langle\hat{P}\Psi|\hat{\Omega}|\hat{P}\Psi\rangle=\langle\Psi|\hat{P}^\dagger\hat{\Omega}\hat{P}|\Psi\rangle\qquad(*)$$

$$\forall\Psi,\Phi\in\mathcal{H}\qquad \langle\Phi|\hat{\Omega}|\Psi\rangle=\frac{1}{4}\Big[\langle\Phi+\Psi|\hat{\Omega}|\Phi+\Psi\rangle+\langle\Phi-\Psi|\hat{\Omega}|\Phi-\Psi\rangle$$
$$-i\langle\Phi+i\Psi|\hat{\Omega}|\Phi+i\Psi\rangle+i\langle\Phi-i\Psi|\hat{\Omega}|\Phi-i\Psi\rangle\Big]$$

类比 $a^*b=\frac{1}{4}\Big[(a+b)^*(a+b)-(a-b)^*(a-b)-i(a+ib)^*(a+ib)+i(a-ib)^*(a-ib)\Big]$

利用（$*$）式，得

$$\langle\Phi|\hat{\Omega}|\Psi\rangle=\frac{1}{4}\Big[\langle\Phi+\Psi|\hat{P}^\dagger\hat{\Omega}\hat{P}|\Phi+\Psi\rangle+\langle\Phi-\Psi|\hat{P}^\dagger\hat{\Omega}\hat{P}|\Phi-\Psi\rangle$$
$$-i\langle\Phi+i\Psi|\hat{P}^\dagger\hat{\Omega}\hat{P}|\Phi+i\Psi\rangle+i\langle\Phi-i\Psi|\hat{P}^\dagger\hat{\Omega}\hat{P}|\Phi-i\Psi\rangle\Big]$$
$$=\langle\Phi|\hat{P}^\dagger\hat{\Omega}\hat{P}|\Psi\rangle\qquad\Longrightarrow\qquad\hat{\Omega}=\hat{P}^\dagger\hat{\Omega}\hat{P}\qquad\Longrightarrow\qquad\hat{P}\hat{\Omega}=\hat{\Omega}\hat{P}$$

证明：算符满足置换对称性

由 $$[\hat{P},\hat{\Omega}]=0$$

可得 $$\hat{\Omega}\hat{P}\Psi(\boldsymbol{x}_1,\boldsymbol{x}_2,\cdots,\boldsymbol{x}_N)=\hat{P}\hat{\Omega}\Psi(\boldsymbol{x}_1,\boldsymbol{x}_2,\cdots,\boldsymbol{x}_N)$$

$$\hat{P}\hat{\Omega}\Psi(\boldsymbol{x}_1,\boldsymbol{x}_2,\cdots,\boldsymbol{x}_N)=\hat{P}\Big(\hat{\Omega}(\boldsymbol{x}_1,\boldsymbol{x}_2,\cdots,\boldsymbol{x}_N)\Psi(\boldsymbol{x}_1,\boldsymbol{x}_2,\cdots,\boldsymbol{x}_N)\Big)$$

$$=\hat{\Omega}(\boldsymbol{x}_{P(1)},\boldsymbol{x}_{P(2)},\cdots,\boldsymbol{x}_{P(N)})\Psi(\boldsymbol{x}_{P(1)},\boldsymbol{x}_{P(2)},\cdots,\boldsymbol{x}_{P(N)})$$

$$\hat{\Omega}\hat{P}\Psi(\boldsymbol{x}_1,\boldsymbol{x}_2,\cdots,\boldsymbol{x}_N)=\hat{\Omega}(\boldsymbol{x}_1,\boldsymbol{x}_2,\cdots,\boldsymbol{x}_N)\Psi(\boldsymbol{x}_{P(1)},\boldsymbol{x}_{P(2)},\cdots,\boldsymbol{x}_{P(N)})$$

由于Ψ是任意的，从而

$$\hat{\Omega}(\boldsymbol{x}_{P(1)},\boldsymbol{x}_{P(2)},\cdots,\boldsymbol{x}_{P(N)})=\Omega(\boldsymbol{x}_1,\boldsymbol{x}_2,\cdots,\boldsymbol{x}_N)$$

例如，多电子哈密顿量

一类特殊的置换(permutation)——交换(transposition)

$$\hat{P}_{jk}\Psi(\boldsymbol{x}_1,\boldsymbol{x}_2,\cdots,\boldsymbol{x}_j,\cdots,\boldsymbol{x}_k,\cdots,\boldsymbol{x}_N)$$

$$\equiv\Psi(\boldsymbol{x}_1,\boldsymbol{x}_2,\cdots,\boldsymbol{x}_k,\cdots,\boldsymbol{x}_j,\cdots,\boldsymbol{x}_N)$$

$$\hat{P}_{jk}^2=\hat{P}_{jk}\hat{P}_{jk}=\hat{I}$$

$$\Longrightarrow\quad \hat{P}_{jk}=\hat{P}_{jk}^{-1}=\hat{P}_{jk}^{\dagger}$$

利用了交换群是二元群

思考题：证明

$$\langle\Psi_A|\hat{\Omega}|\Psi_S\rangle=0$$

算符的本征态要么全都对称，要么全都反对称

交换算符是厄密算符！

厄密算符有本征态，且本征值为实数，令

$$\hat{P}_{jk}\Psi=a_{jk}\Psi\quad\Longrightarrow\quad\hat{P}_{jk}^2\Psi=a_{jk}^2\Psi=\Psi$$

$$\Longrightarrow a_{jk}=\pm1$$

玻色子 $\hat{P}\Psi_S=\Psi_S,\quad\forall\hat{P}$

费米子 $\hat{P}_{jk}\Psi_A=-\Psi_A,\quad\forall\hat{P}_{jk}$

Pauli原理

三　多电子原子

以氦原子为例，2个电子

$$\hat{H}=\overbrace{-\frac{\hbar^2}{2m_e}\nabla_1^2-\frac{2e^2}{4\pi\varepsilon_0 r_1}}^{\hat{H}_1}\overbrace{-\frac{\hbar^2}{2m_e}\nabla_2^2-\frac{2e^2}{4\pi\varepsilon_0 r_2}}^{\hat{H}_2}+\frac{e^2}{4\pi\varepsilon_0 r_{12}}$$

哈密顿量中含有单电子算符和双电子算符

假如电子间没有相互作用，则可以分离变量

真实情况电子间有相互作用，以下变量分离不成立，而是**单电子(轨道)近似**！

$$\Psi(\boldsymbol{r}_1,\boldsymbol{r}_2)=\psi_1(\boldsymbol{r}_1)\psi_2(\boldsymbol{r}_2)$$

$$\hat{H}_1\psi_1(\boldsymbol{r}_1)=E_1\psi_1(\boldsymbol{r}_1)$$

$$\hat{H}_2\psi_2(\boldsymbol{r}_2)=E_2\psi_2(\boldsymbol{r}_2)$$

$$(\hat{H}_1+\hat{H}_2)\Psi(\boldsymbol{r}_1,\boldsymbol{r}_2)=(E_1+E_2)\Psi(\boldsymbol{r}_1,\boldsymbol{r}_2)$$

注意到 $\hat{H}_1=\hat{h}(\boldsymbol{r}_1)$ $\hat{H}_2=\hat{h}(\boldsymbol{r}_2)$

令 $\tilde{\Psi}(\boldsymbol{r}_1,\boldsymbol{r}_2)=c_1\psi_1(\boldsymbol{r}_1)\psi_2(\boldsymbol{r}_2)+c_2\psi_1(\boldsymbol{r}_2)\psi_2(\boldsymbol{r}_1)$

有 $(\hat{H}_1+\hat{H}_2)\tilde{\Psi}(\boldsymbol{r}_1,\boldsymbol{r}_2)=(E_1+E_2)\tilde{\Psi}(\boldsymbol{r}_1,\boldsymbol{r}_2)$

考虑对称性可取 $c_1=\pm c_2$

单电子近似: $\Psi(\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots)=\psi_1(\boldsymbol{r}_1)\psi_2(\boldsymbol{r}_2)\cdots$

$$\Psi^{\mathrm{sym}}(\boldsymbol{r}_1,\boldsymbol{r}_2,\cdots)=\hat{S}\Big(\psi_1(\boldsymbol{r}_1)\psi_2(\boldsymbol{r}_2)\cdots\Big)$$

泡利不相容原理

考虑双电子问题 $\Psi(2,1) = -\Psi(1,2)$

对空间与自旋进行分离

自旋波函数：四种可能状态 $\alpha(1)\alpha(2)$ $\beta(1)\beta(2)$ $\alpha(1)\beta(2)$ $\alpha(2)\beta(1)$

后两种状态需进行对称性重组

$$\sigma_+(1,2) = \left(\frac{1}{2^{1/2}}\right)\{\alpha(1)\beta(2) + \beta(1)\alpha(2)\}$$

$$\sigma_-(1,2) = \left(\frac{1}{2^{1/2}}\right)\{\alpha(1)\beta(2) - \beta(1)\alpha(2)\}$$

基态空间波函数：考虑轨道近似，占据1s轨道

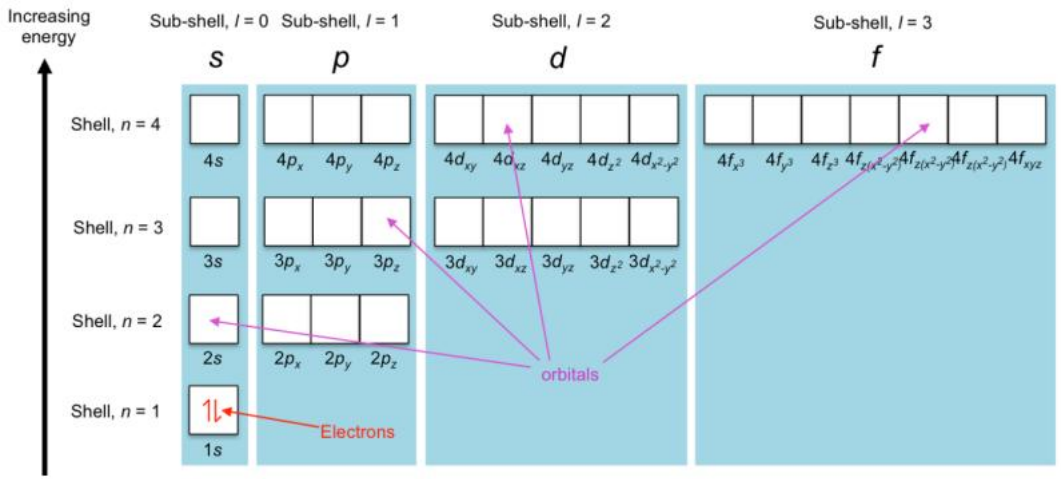
基态全波函数 $\psi(1)\psi(2)\alpha(1)\alpha(2)$ $\psi(1)\psi(2)\beta(1)\beta(2)$

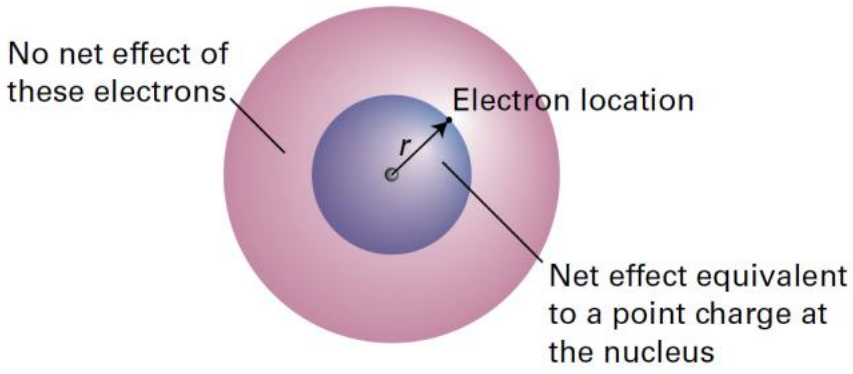
$\psi(1)\psi(2)\sigma_+(1,2)$ $\psi(1)\psi(2)\sigma_-(1,2)$

只有最后一个可取
(He原子电子构型)

思考：激发态的全波函数

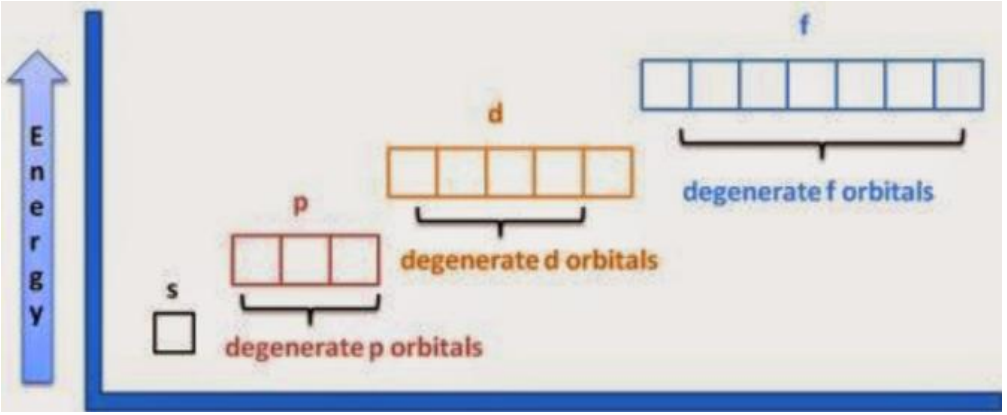
单电子原子轨道：spdf亚层简并

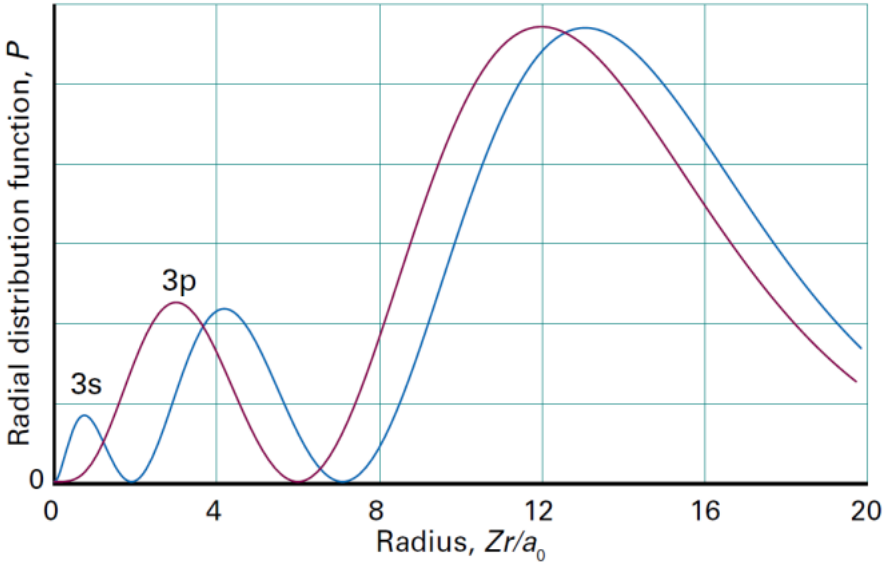




单电子图像中的**内层电子屏蔽**

多电子原子轨道：亚层简并破除





http://monomole.com/2019/11/16/intermediate-quantum-chemistry-2/

不同亚层轨道的钻穿效应

屏蔽常数

$$Z_{\text{eff}} = Z - \sigma$$

同层轨道能量

$$s < p < d < f$$

不同层轨道能量

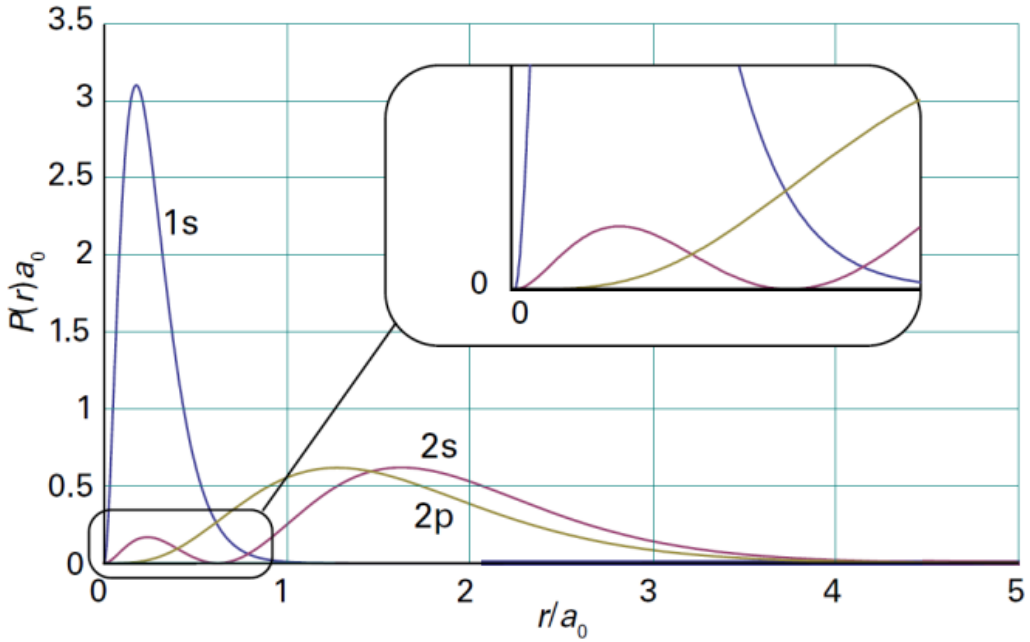
$$1s < 2s < 2p < 3s < 3p$$

$$< 4s < 3d < 4p < 5s < 4d < 5p < 6s$$

Aufbau规则

注意：以上都基于
单电子近似图像

Element	Z	Orbital	Z_{eff}
He	2	1s	1.6875
C	6	1s	5.6727
		2s	3.2166
		2p	3.1358



C**原子轨道**的径向分布函数

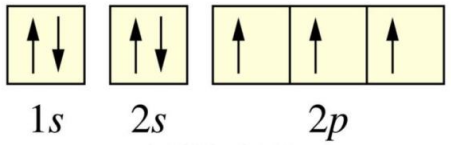
洪特(Hund)规则：

- 关于**亚层**：电子配对之前优先占据不同亚层空间轨道
  (库伦排斥小)

$$[He]2s^2 2p_x^1 2p_y^1 2p_z^1 \quad [He]2s^2 2p_x^2 2p_y^1 2p_z^1$$

- 关于**自旋**：原子的基态采取未配对电子最多的构型
  (自旋关联能)

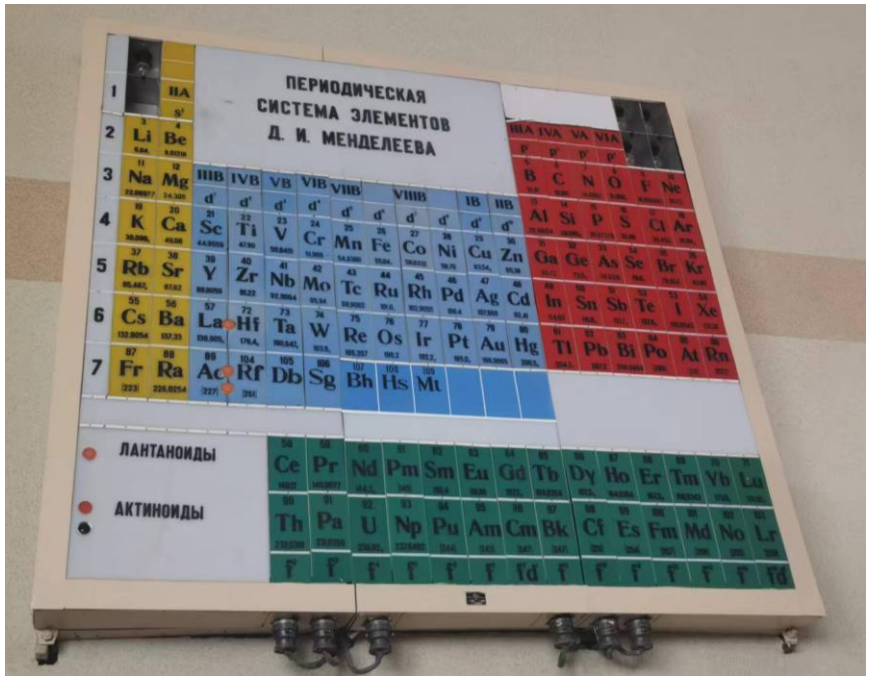
N:

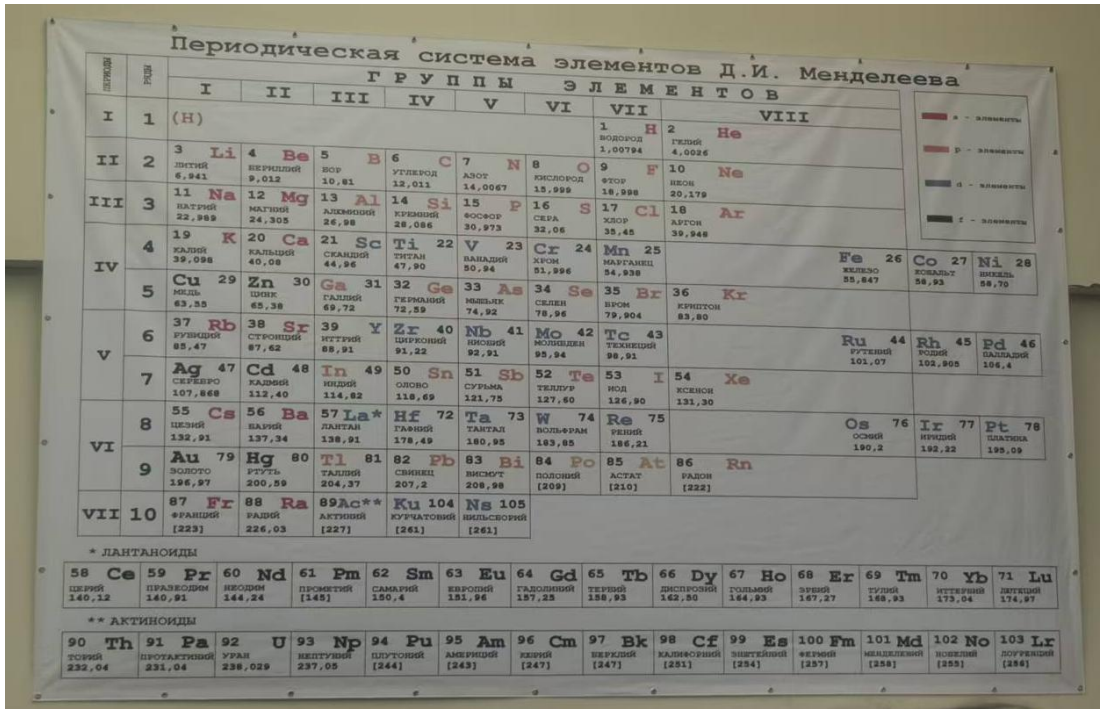


自旋多重度最大

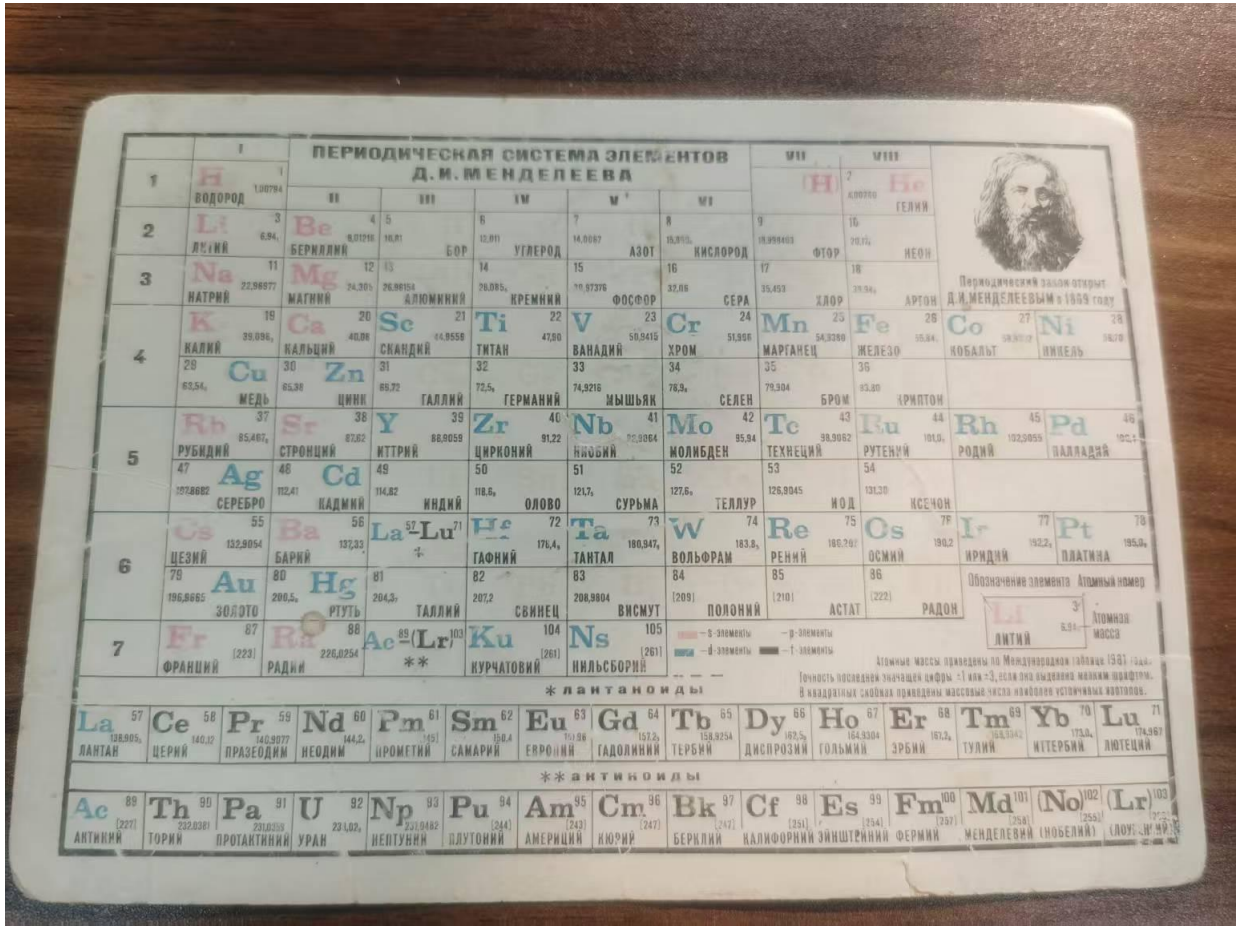
https://chem.libretexts.org/

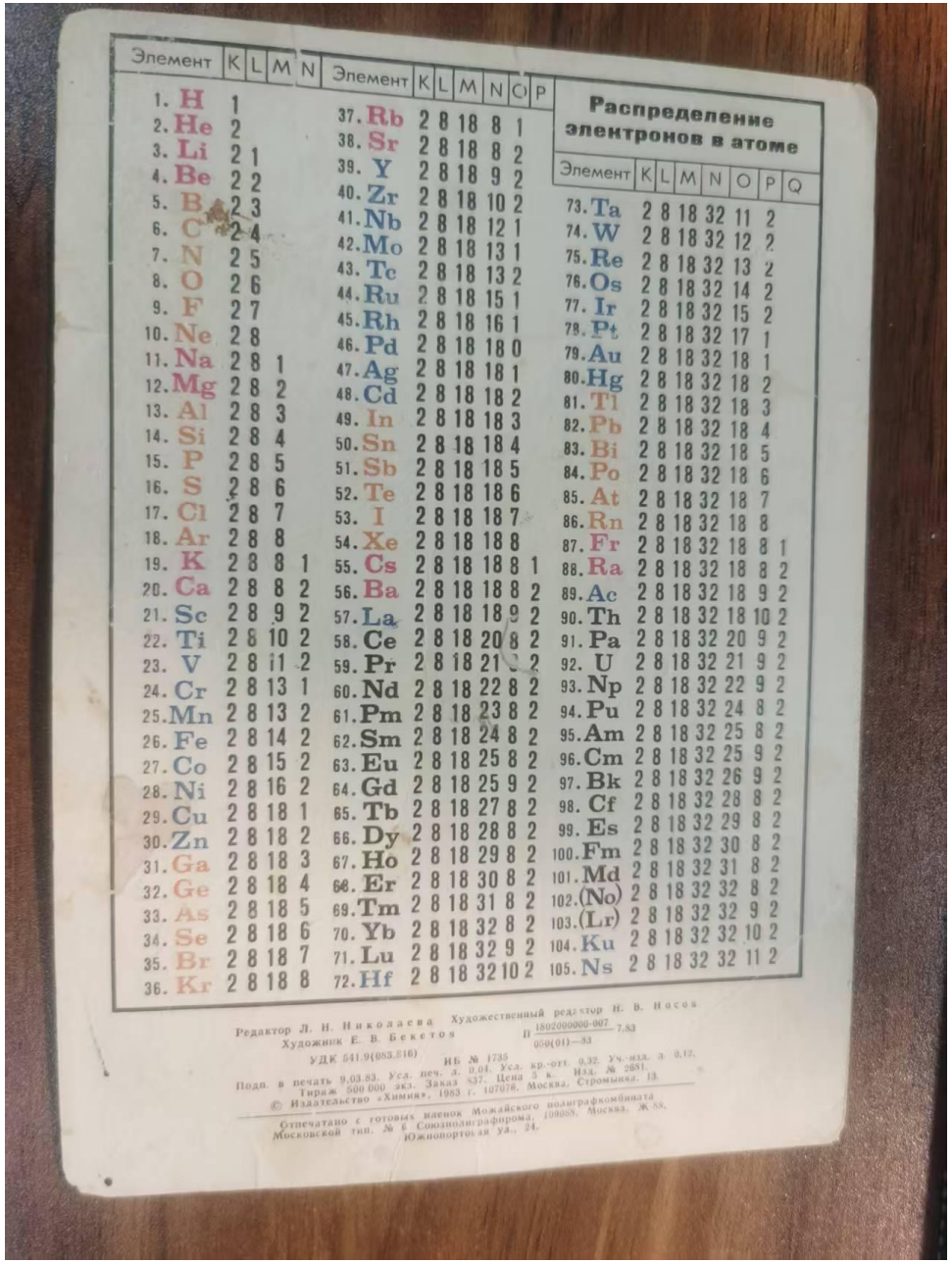
Atomic number	Symbol	Electron configuration	Atomic number	Symbol	Electron configuration	Atomic number	Symbol	Electron configuration
1	H	$1s^1$	37	Rb	$[Kr]5s^1$	73	Ta	$[Xe]6s^2 4f^{14} 5d^3$
2	He	$1s^2$	38	Sr	$[Kr]5s^2$	74	W	$[Xe]6s^2 4f^{14} 5d^4$
3	Li	$[He]2s^1$	39	Y	$[Kr]5s^2 4d^1$	75	Re	$[Xe]6s^2 4f^{14} 5d^5$
4	Be	$[He]2s^2$	40	Zr	$[Kr]5s^2 4d^2$	76	Os	$[Xe]6s^2 4f^{14} 5d^6$
5	B	$[He]2s^2 2p^1$	41	Nb	$[Kr]5s^1 4d^4$	77	Ir	$[Xe]6s^2 4f^{14} 5d^7$
6	C	$[He]2s^2 2p^2$	42	Mo	$[Kr]5s^1 4d^5$	78	Pt	$[Xe]6s^1 4f^{14} 5d^9$
7	N	$[He]2s^2 2p^3$	43	Tc	$[Kr]5s^2 4d^5$	79	Au	$[Xe]6s^1 4f^{14} 5d^{10}$
8	O	$[He]2s^2 2p^4$	44	Ru	$[Kr]5s^1 4d^7$	80	Hg	$[Xe]6s^2 4f^{14} 5d^{10}$
9	F	$[He]2s^2 2p^5$	45	Rh	$[Kr]5s^1 4d^8$	81	Tl	$[Xe]6s^2 4f^{14} 5d^{10} 6p^1$
10	Ne	$[He]2s^2 2p^6$	46	Pd	$[Kr]4d^{10}$	82	Pb	$[Xe]6s^2 4f^{14} 5d^{10} 6p^2$
11	Na	$[Ne]3s^1$	47	Ag	$[Kr]5s^1 4d^{10}$	83	Bi	$[Xe]6s^2 4f^{14} 5d^{10} 6p^3$
12	Mg	$[Ne]3s^2$	48	Cd	$[Kr]5s^2 4d^{10}$	84	Po	$[Xe]6s^2 4f^{14} 5d^{10} 6p^4$
13	Al	$[Ne]3s^2 3p^1$	49	In	$[Kr]5s^2 4d^{10} 5p^1$	85	At	$[Xe]6s^2 4f^{14} 5d^{10} 6p^5$
14	Si	$[Ne]3s^2 3p^2$	50	Sn	$[Kr]5s^2 4d^{10} 5p^2$	86	Rn	$[Xe]6s^2 4f^{14} 5d^{10} 6p^6$
15	P	$[Ne]3s^2 3p^3$	51	Sb	$[Kr]5s^2 4d^{10} 5p^3$	87	Fr	$[Rn]7s^1$
16	S	$[Ne]3s^2 3p^4$	52	Te	$[Kr]5s^2 4d^{10} 5p^4$	88	Ra	$[Rn]7s^2$
17	Cl	$[Ne]3s^2 3p^5$	53	I	$[Kr]5s^2 4d^{10} 5p^5$	89	Ac	$[Rn]7s^2 6d^1$
18	Ar	$[Ne]3s^2 3p^6$	54	Xe	$[Kr]5s^2 4d^{10} 5p^6$	90	Th	$[Rn]7s^2 6d^2$
19	K	$[Ar]4s^1$	55	Cs	$[Xe]6s^1$	91	Pa	$[Rn]7s^2 5f^2 6d^1$
20	Ca	$[Ar]4s^2$	56	Ba	$[Xe]6s^2$	92	U	$[Rn]7s^2 5f^3 6d^1$
21	Sc	$[Ar]4s^2 3d^1$	57	La	$[Xe]6s^2 5d^1$	93	Np	$[Rn]7s^2 5f^4 6d^1$
22	Ti	$[Ar]4s^2 3d^2$	58	Ce	$[Xe]6s^2 4f^1 5d^1$	94	Pu	$[Rn]7s^2 5f^6$
23	V	$[Ar]4s^2 3d^3$	59	Pr	$[Xe]6s^2 4f^3$	95	Am	$[Rn]7s^2 5f^7$
24	Cr	$[Ar]4s^1 3d^5$	60	Nd	$[Xe]6s^2 4f^4$	96	Cm	$[Rn]7s^2 5f^7 6d^1$
25	Mn	$[Ar]4s^2 3d^5$	61	Pm	$[Xe]6s^2 4f^5$	97	Bk	$[Rn]7s^2 5f^9$
26	Fe	$[Ar]4s^2 3d^6$	62	Sm	$[Xe]6s^2 4f^6$	98	Cf	$[Rn]7s^2 5f^{10}$
27	Co	$[Ar]4s^2 3d^7$	63	Eu	$[Xe]6s^2 4f^7$	99	Es	$[Rn]7s^2 5f^{11}$
28	Ni	$[Ar]4s^2 3d^8$	64	Gd	$[Xe]6s^2 4f^7 5d^1$	100	Fm	$[Rn]7s^2 5f^{12}$
29	Cu	$[Ar]4s^1 3d^{10}$	65	Tb	$[Xe]6s^2 4f^9$	101	Md	$[Rn]7s^2 5f^{13}$
30	Zn	$[Ar]4s^2 3d^{10}$	66	Dy	$[Xe]6s^2 4f^{10}$	102	No	$[Rn]7s^2 5f^{14}$
31	Ga	$[Ar]4s^2 3d^{10} 4p^1$	67	Ho	$[Xe]6s^2 4f^{11}$	103	Lr	$[Rn]7s^2 5f^{14} 6d^1$
32	Ge	$[Ar]4s^2 3d^{10} 4p^2$	68	Er	$[Xe]6s^2 4f^{12}$	104	Rf	$[Rn]7s^2 5f^{14} 6d^2$
33	As	$[Ar]4s^2 3d^{10} 4p^3$	69	Tm	$[Xe]6s^2 4f^{13}$	105	Db	$[Rn]7s^2 5f^{14} 6d^3$
34	Se	$[Ar]4s^2 3d^{10} 4p^4$	70	Yb	$[Xe]6s^2 4f^{14}$	106	Sg	$[Rn]7s^2 5f^{14} 6d^4$
35	Br	$[Ar]4s^2 3d^{10} 4p^5$	71	Lu	$[Xe]6s^2 4f^{14} 5d^1$	107	Bh	$[Rn]7s^2 5f^{14} 6d^5$
36	Kr	$[Ar]4s^2 3d^{10} 4p^6$	72	Hf	$[Xe]6s^2 4f^{14} 5d^2$	108	Hs	$[Rn]7s^2 5f^{14} 6d^6$
						109	Mt	$[Rn]7s^2 5f^{14} 6d^7$
						110	Ds	$[Rn]7s^1 5f^{14} 6d^9$
						111	Rg	$[Rn]7s^1 5f^{14} 6d^{10}$





门捷列夫元素周期表（初版）





Элемент	K	L	M	N	Элемент	K	L	M	N	O	P
1. H	1				37. Rb	2	8	18	8	1	
2. He	2				38. Sr	2	8	18	8	2	
3. Li	2	1			39. Y	2	8	18	9	2	
4. Be	2	2			40. Zr	2	8	18	10	2	
5. B	2	3			41. Nb	2	8	18	12	1	
6. C	2	4			42. Mo	2	8	18	13	1	
7. N	2	5			43. Tc	2	8	18	13	2	
8. O	2	6			44. Ru	2	8	18	15	1	
9. F	2	7			45. Rh	2	8	18	16	1	
10. Ne	2	8			46. Pd	2	8	18	18	0	
11. Na	2	8	1		47. Ag	2	8	18	18	1	
12. Mg	2	8	2		48. Cd	2	8	18	18	2	
13. Al	2	8	3		49. In	2	8	18	18	3	
14. Si	2	8	4		50. Sn	2	8	18	18	4	
15. P	2	8	5		51. Sb	2	8	18	18	5	
16. S	2	8	6		52. Te	2	8	18	18	6	
17. Cl	2	8	7		53. I	2	8	18	18	7	
18. Ar	2	8	8		54. Xe	2	8	18	18	8	
19. K	2	8	8	1	55. Cs	2	8	18	18	8	1
20. Ca	2	8	8	2	56. Ba	2	8	18	18	8	2
21. Sc	2	8	9	2	57. La	2	8	18	18	9	2
22. Ti	2	8	10	2	58. Ce	2	8	18	20	8	2
23. V	2	8	11	2	59. Pr	2	8	18	21	8	2
24. Cr	2	8	13	1	60. Nd	2	8	18	22	8	2
25. Mn	2	8	13	2	61. Pm	2	8	18	23	8	2
26. Fe	2	8	14	2	62. Sm	2	8	18	24	8	2
27. Co	2	8	15	2	63. Eu	2	8	18	25	8	2
28. Ni	2	8	16	2	64. Gd	2	8	18	25	9	2
29. Cu	2	8	18	1	65. Tb	2	8	18	27	8	2
30. Zn	2	8	18	2	66. Dy	2	8	18	28	8	2
31. Ga	2	8	18	3	67. Ho	2	8	18	29	8	2
32. Ge	2	8	18	4	68. Er	2	8	18	30	8	2
33. As	2	8	18	5	69. Tm	2	8	18	31	8	2
34. Se	2	8	18	6	70. Yb	2	8	18	32	8	2
35. Br	2	8	18	7	71. Lu	2	8	18	32	9	2
36. Kr	2	8	18	8	72. Hf	2	8	18	32	10	2

Распределение электронов в атоме

Элемент	K	L	M	N	O	P	Q
73. Ta	2	8	18	32	11	2	
74. W	2	8	18	32	12	2	
75. Re	2	8	18	32	13	2	
76. Os	2	8	18	32	14	2	
77. Ir	2	8	18	32	15	2	
78. Pt	2	8	18	32	17	1	
79. Au	2	8	18	32	18	1	
80. Hg	2	8	18	32	18	2	
81. Tl	2	8	18	32	18	3	
82. Pb	2	8	18	32	18	4	
83. Bi	2	8	18	32	18	5	
84. Po	2	8	18	32	18	6	
85. At	2	8	18	32	18	7	
86. Rn	2	8	18	32	18	8	
87. Fr	2	8	18	32	18	8	1
88. Ra	2	8	18	32	18	8	2
89. Ac	2	8	18	32	18	9	2
90. Th	2	8	18	32	18	10	2
91. Pa	2	8	18	32	20	9	2
92. U	2	8	18	32	21	9	2
93. Np	2	8	18	32	22	9	2
94. Pu	2	8	18	32	24	8	2
95. Am	2	8	18	32	25	8	2
96. Cm	2	8	18	32	25	9	2
97. Bk	2	8	18	32	26	9	2
98. Cf	2	8	18	32	28	8	2
99. Es	2	8	18	32	29	8	2
100. Fm	2	8	18	32	30	8	2
101. Md	2	8	18	32	31	8	2
102. (No)	2	8	18	32	32	8	2
103. (Lr)	2	8	18	32	32	9	2
104. Ku	2	8	18	32	32	10	2
105. Ns	2	8	18	32	32	11	2

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自旋关联

以两个轨道为例，满足对称性的空间波函数为

$$\Psi_{\pm} = \left(\frac{1}{2^{1/2}}\right)\{\psi_a(\boldsymbol{r}_1)\psi_b(\boldsymbol{r}_2) \pm \psi_b(\boldsymbol{r}_1)\psi_a(\boldsymbol{r}_2)\}$$

思考：为什么He原子基态是单重态，不是三重态？

空间波函数**对称**，则自旋波函数**反对称**

空间波函数**反对称**，则自旋波函数**对称**

空间波函数**反对称**时 $\Psi_{-}(\boldsymbol{r},\boldsymbol{r}) = 0$ 电子间库伦排除小，能量低

从而，基态自旋**对称** $\alpha(1)\alpha(2)$ $\beta(1)\beta(2)$ $\sigma_{+}(1,2) = \left(\frac{1}{2^{1/2}}\right)\{\alpha(1)\beta(2) + \beta(1)\alpha(2)\}$

这三个态能量简并，是三重态，例如O原子

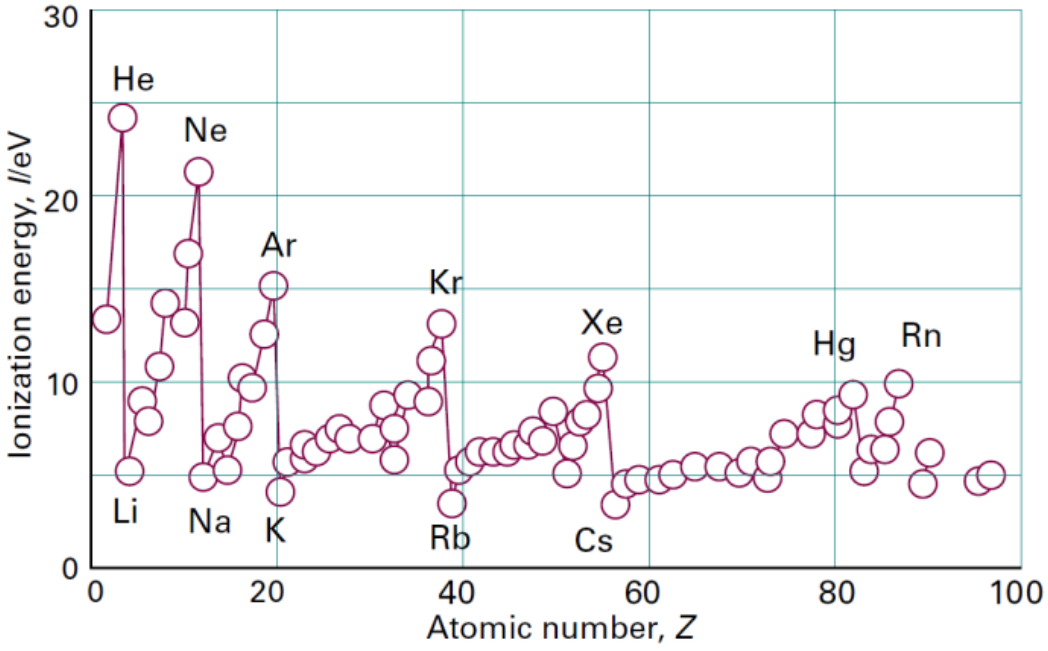
注意：第三个态自旋反平行

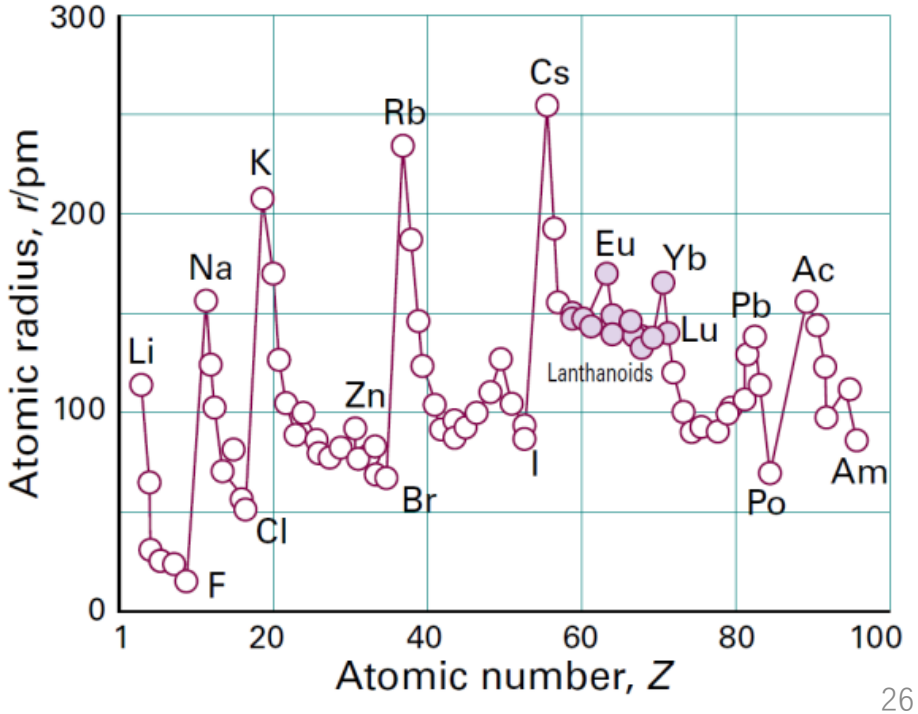
以上单电子图像有很多反例，如第一过渡周期 $[\mathrm{Ar}]3d^n4s^2$

元素周期律

- 原子半径
- 镧系收缩
- 电离能(第一、第二、……)
- 电子亲合能

Li 157	Be 112	B 88	C 77	N 74	O 66	F 64
Na 191	Mg 160	Al 143	Si 118	P 110	S 104	Cl 99
K 235	Ca 197	Ga 153	Ge 122	As 121	Se 117	Br 114
Rb 250	Sr 215	In 167	Sn 158	Sb 141	Te 137	I 133
Cs 272	Ba 224	Tl 171	Pb 175	Bi 182	Po 167	





基于单电子图像（引入轨道）

中心力场法（central force field)

Hartree-Fock自洽场法（self-consistent field）

密度泛函方法（density functional theory)

多体微扰理论（many-body perturbation theory）

求解全波函数

组态相互作用方法（configuration interaction）

耦合簇方法（coupled cluster）

量子蒙特卡洛法（quantum Monte Carlo）